Q2:- Compute the Riemann sum using 20 equally spaced sub-intervals for the following definite integral

$$\int_0^2 e^{x^2}\,dx.$$

Write also the code.

Q3:- $A = \begin{bmatrix} 2 & 3 & 4 & 7 \\ 3 & 2 & 1 & 8 \\ 7 & 8 & 9 & 10 \\ 4 & 5 & 6 & 2 \end{bmatrix}$

→ Determine square matrix whose columns represent the Eigenvectors of the matrix A.

→ Determine the diagonal matrix having the Eigenvalues of the matrix A on its main diagonal.

Q4.- Compute the following relation for $i = 0:10$, and choosing the values of the constants $c = 1$ and $d = 2$.

$$\Omega_i^{(c,d)} = \frac{4^{c+d+10}\,\Gamma(i+5c+d+10)\,(i+19)!}{(2i+c+d+12)\,\Gamma(i+c+5)\,(i+10c+d+15)!}$$

Q5.- Compute the following relation for $k = 0:10$, and choosing the values of the constants $a = 1$ and $b = 9$.

$$\Lambda_k^{(a,b)} = \frac{2^{(a+b+15)}\,\Gamma(k+3a+10b+10)\,(k+4+a+b)!\,\Gamma(a+b+5)}{(3k+a+b+4)\,\Gamma(k+a+4b+14)\,(k+10a+15b+12)!}$$

Forwarded

1)Use ode45 to plot the problem
2)eucler improved formula code also plot
3)trapezoidal rule question
4)lagender polynomial function code
5)omega polynomial function 14 lecture s
Had question 10 marks ka tha

8:20 pm

Solve the equation with solve Command

① $2x+2y=3, \; 3x^2+2y=4$

Also plot the graph of $3x^2+2y=4$

If $A = \begin{pmatrix} 2 & 3 & 4 & 7 \\ 3 & 2 & 1 & 8 \\ 7 & 8 & 9 & 10 \\ 4 & 5 & 6 & 2 \end{pmatrix}$

②

- To find the Square and Eigen Vector of A
- To determine the diagonal matrix of A with Eigen Values.

---

③

Using Simpson to Solve $Y(x) = \sqrt{(1+x^3)}$ three - eight Method

having interval [1,5], Sub Interval <u>30</u>

---

④ If $i = 0:10$ and $c = 1$, $d = 2$

Then find the value of

$$\Omega_i(c,d) = \frac{4^{c+d+10}\,\Gamma(i+5c+d+10)\,(i+19)!}{(2i+c+d+12)\,\Gamma(i+c+5)\,(i+10c+d+15)!}$$

⑤ If $k = 0:10$, $a=1$, $b=1$

Then Compute the value of

$$\Delta_k(a,b) = \frac{2^{a+b+15}\,\Gamma(k+3a+10b+10)^{(k+4+a+b)}}{(3k+a+b+4)\,\Gamma(k+a+b+4)(k+10a+15\ldots}$$

MTH - 643 - Final - Paper

Q1= Find the nomerical solution of following differential equation on the interval $[0, 2]$ using Taylor's method of the second interval for $n = 10$.

$$\frac{dy}{dx} = \cos(xy) \quad , \quad y(0) = \pi$$

Note. Iteration releaxation of Taylor's method is;

$$y_{k+1} \approx y_k + f(x_k, y_k)\Delta x + \left(\frac{\partial f}{\partial x}(x_k, y_k) + \frac{\partial f}{\partial y}(x_k, y_k)\right)$$

Write code in script file with m-extention in MATLAB.

Construct the one–dimensional Orthogonal Shifted Jacobi function vector, $P_n^{(\alpha,\beta)}(x)$ defined on $[0,1]$ for $i = 0 : \cancel{19}\ 24$, and by choosing the values of the parameters $\alpha = \cancel{0}\ 1$, and $\beta = \cancel{0}\ 1$ using the following relation:

$$P_n^{(\alpha,\beta)}(x) = \sum_{k=0}^{i} \frac{(-1)^{(i-k)}\Gamma(i+\beta+1)\Gamma(i+k+\alpha+\beta+1)}{\Gamma(k+\beta+1)\Gamma(i+\alpha+\beta+1)\Gamma(i-k+1)\Gamma(k+1)1^k} x^k,$$

Copied to clipboard.

Compute the following relation for j=0:20, and choosing the values of the constants $a = 1$, and $b = 2$:

$$\Omega_j^{(a,b)} = \frac{4^{a+b+1}\Gamma(j+3a+4b+1)(j+9)!}{(2j+a+b+1)\Gamma(j+a+5)(j+10a+b+1)!}.$$

- Compute the definite integral of $y(x) = \sqrt{(1 + x^3)}$ defined on the interval $[1, 5]$ using Simpson's 3/8 ~~One Third~~ Rule by dividing the interval $[1, 5]$ into ~~100~~ 30 sub-intervals of uniform length.

Note:
The general form of Simpson's 3/8 ~~One Third~~ Rule is given as under:

$$\int_{x_0}^{x_{n+h}} y(x)dx \approx \frac{\Delta}{3}\Big[(y(x_0) + y(x_n)) + 4\,(y(x_1) + y(x_3) + \ldots + y(x_{n-1}))$$
$$+ 2\,(y(x_2) + y(x_4) + \ldots + y(x_{n-2}))\Big].$$

1) solve system of ordinary differential equations + plot graph (lec#7)
2) trapezoidal rule (lec#12)
3) solve eq using ode45 command (lec#9)
4)Orthogonal Shifted Jacobi Polynomials(lec#13)
5) question similar to lec#15 lastq

3:22 pm

Matlab (2019)

Q1

plot family of curve by ode45 on [-2, 2]

$y'(x) = e^{x^2}$

Q2 Euler improved formula

Lecture # 8

$f = \dfrac{y-1}{y+1}$

Q3 :- Trapozoidal Rule (lecture # 12)

Q4 Lecture # 15 (Pg No 17)

$\Rightarrow$ (One - dimentional orthognal shifte Jacobi)

Q5 Lecture # 14

orthognal shifted legendre

MATLAB 7.10.0 (R2010a)

Edit Text Go Cell Tools Debug Parallel Desktop Window Help

Current Folder: F:\vu\4rt semester\mth643\practice\final part

hortcuts How to Add What's New

Editor - C:\Users\mianw\AppData\Local\Temp\Rar$DIa15880.4250\onedimensionalOrthogonalShiftedJacobi.m

Stack: Base

```
%parameters ? = 0, and ? = 0 using the following relation:
function P=onedimensionalOrthogonalShiftedJacobi(M,T,alpha,beta)
close all;
clear all;
clc;
syms M T k x
M=20;
T=1;
alpha=0;
beta=0;
for i=0:M-1
    a1=(-1)^(i-k);
    a2=gamma(i+beta+1);
    a3=gamma(i+k+alpha+beta+1);
    d1=gamma(k+beta+1);
    d2=gamma(i+alpha+beta+1);
    d3=gamma(i-k+1);
    d4=gamma(k+1);
    d5=T^k;
    aa=(a1*a2*a3)/(d1*d2*d3*d4*d5);
    P(i+1)=symsum(aa*x^k,k,0,i);
end
P=transpose(P)
```

ample.m lec_13_4th_Example.m orthogonal1.m orthogonal2.m orthogonal3.m orthogonal4.m lec1

Start

Type here to search

- Consider the following example.

Example:

- Compute the eigen values and eigen vectors for the following matrix:

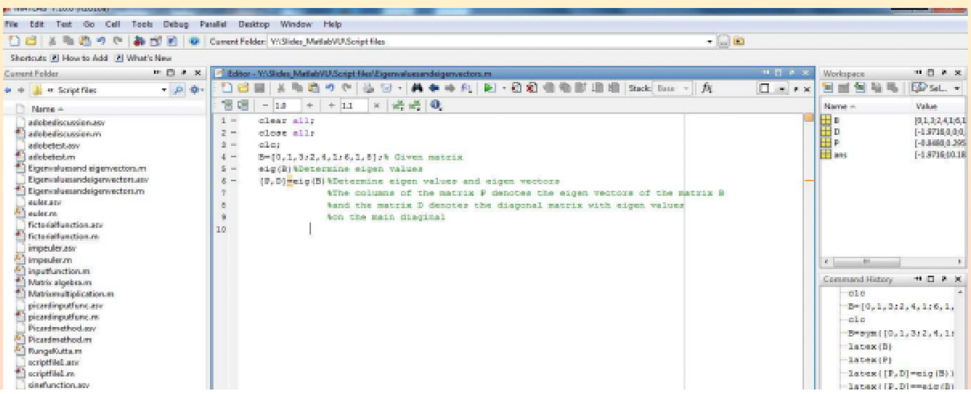
$$B=\begin{pmatrix} 0 & 1 & 3 \\ 2 & 4 & 1 \\ 6 & 1 & 8 \end{pmatrix}$$

- The recipe of the solution is as under:
- B=[0,1,3;2,4,1;6,1,8];
  eig(B)
  $[P, D] = eig(B)$

Note:

- The columns of the matrix P denotes the eigen vectors of the matrix B.
- The matrix D denotes the diagonal matrix having eigen values on the main diaginal.

- Computation of the eigen values and eigen vectors.



Lec11ExQ02 - Notepad

File Edit Format View Help

```
% Lecture # 11, Exercise Q.2
% Consider the system of equations x1+3x3+x5=-1, 4x1+x2+x3+6x5=1,
% -3x1-3x2-3x3+x4-19x5=6, 10x1-4x2+38x3+2x4-12x5=0
%Enter into MatLab as an augmented matrix named Ab
% Use elementary row operations to reduce it to reduced  echelon form
% Give the solution of the system
clear all
clc
A =[1 0 3 0 1;4 1 1 0 6;-3 -3 -3 1 -19;10 -4 38 2 -12]
b = [-1;1;6;0]
Ab = [A b]
Ab(2,:)=Ab(2,:)-4*Ab(1,:);
Ab(3,:)=Ab(3,:)+3*Ab(1,:);
Ab(4,:)=Ab(4,:)-10*Ab(1,:);
Ab(3,:)=Ab(3,:)+3*Ab(2,:);
Ab(4,:)=Ab(4,:)+4*Ab(2,:);
Ab(3,:)=(-1/27)*Ab(3,:);
Ab(1,:)=Ab(1,:)-3*Ab(3,:);
Ab(2,:)=Ab(2,:)+11*Ab(3,:);
Ab(4,:)=Ab(4,:)+36*Ab(3,:);
Ab(4,:)=(1/0.6667)*Ab(4,:);
Ab(1,:)=Ab(1,:)-(0.1111)*Ab(4,:);
Ab(2,:)=Ab(2,:)+(0.4074)*Ab(4,:);
Ab(3,:)=Ab(3,:)+(0.0370)*Ab(4,:);
Ab
disp('x5 is free variable, let x5=1 then other values are as under');
X = Ab(:,6)-Ab(:,5);
x1 =1
x2 =X(1,:)
x3 = X(2,:)
x4 = X(3,:)
x5 = X(4,:)
```

Lec11ExQ02 - Notepad

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```
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